



ESTATE MIND

Group 5 - 4 DS 5

Team



Nourhen Mraeih



Wissal Bahar



Salma Alaya



Manar Zaghouani



Yosser Ben
Mahmoud



Wissem Bahar

Overview

- I. Project Context
- II. Problem Statement & Target Users
- III. Expert Validation & Field Insights
- IV. Market Analysis & Value Proposition
- V. Business Objectives & Data Science Objectives
- VI. Key Performance Indicators (KPIs)
- VII. TDSP Methodology & Team Roles
- VIII. Ethics, Sustainability & Responsible AI
- IX. Next Steps: Data Understanding & Acquisition



01. Project Context

Context

Real estate is a **strategic sector** in Tunisia, impacting households, investors and public institutions.

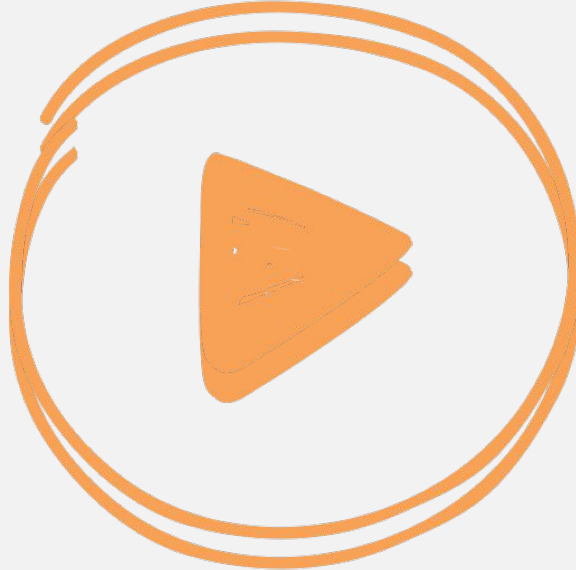
Globally, **PropTech** platforms are transforming the industry through **data** and **AI**.

In **Tunisia**, data-driven real-estate solutions remain **limited**, creating strong **opportunities** for **innovation**.



02. Problem statement & Target users

Problem Statement & Target Users



Lien vidéo:

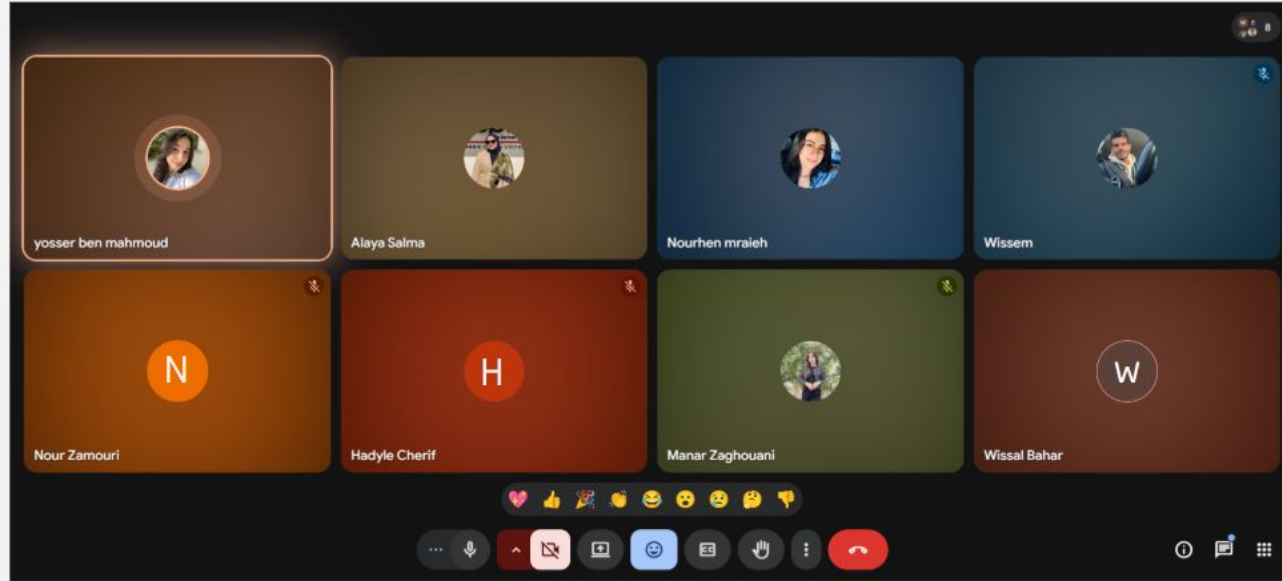
https://www.youtube.com/watch?v=okQdGu_0WAI



In Tunisia, the real-estate market is booming,

03. Expert Validation & Field Insights

Real-World Investigation





— Ahmed Rebai, Data Scientist
and Head of AI Delivery at
Value



— Nour Zamouri, a field
agent in a real-estate
agency in Ain Zaghouan,



— Hane Alaya, owner of
the agency Ma-Vie-Là,
responsible for visits
and contracts

04. Market Analysis & Value Proposition

Market Analysis



Strengths



- Automated price estimation
- Advanced statistical models
- Large transaction datasets

Limits



- Relies on highly structured public records
- Requires complete and centralized data
- Limited fraud detection

Market Analysis



idealista

Strengths



- Interactive maps
- Heatmaps by district
- Territorial price analysis

Limits



- Focus mainly on visualization
- Limited verification of listings
- No legal compliance layer

Market Analysis



Strengths



- Dashboards
- Investor-oriented analytics
- Strategic macro-level insights

Limits



- Oriented toward reports rather than transaction-level decisions
- Limited personalization for buyers and sellers
- No integrated legal compliance engine

Market Analysis



PropertyGuru

Strengths



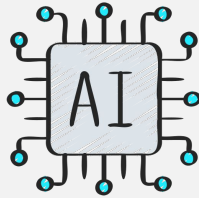
- Dashboards
- Investor-oriented analytics
- Strategic macro-level insights

Limits



- Oriented toward reports rather than transaction-level decisions
- Limited personalization for buyers and sellers
- No integrated legal compliance engine

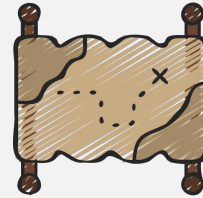
Added Value



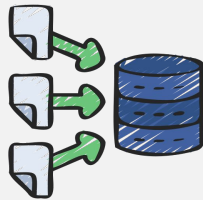
Agentic AI
Architecture



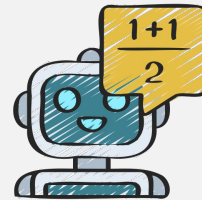
RegTech Layer for
Real Estate



Territorial & Macro
Intelligence



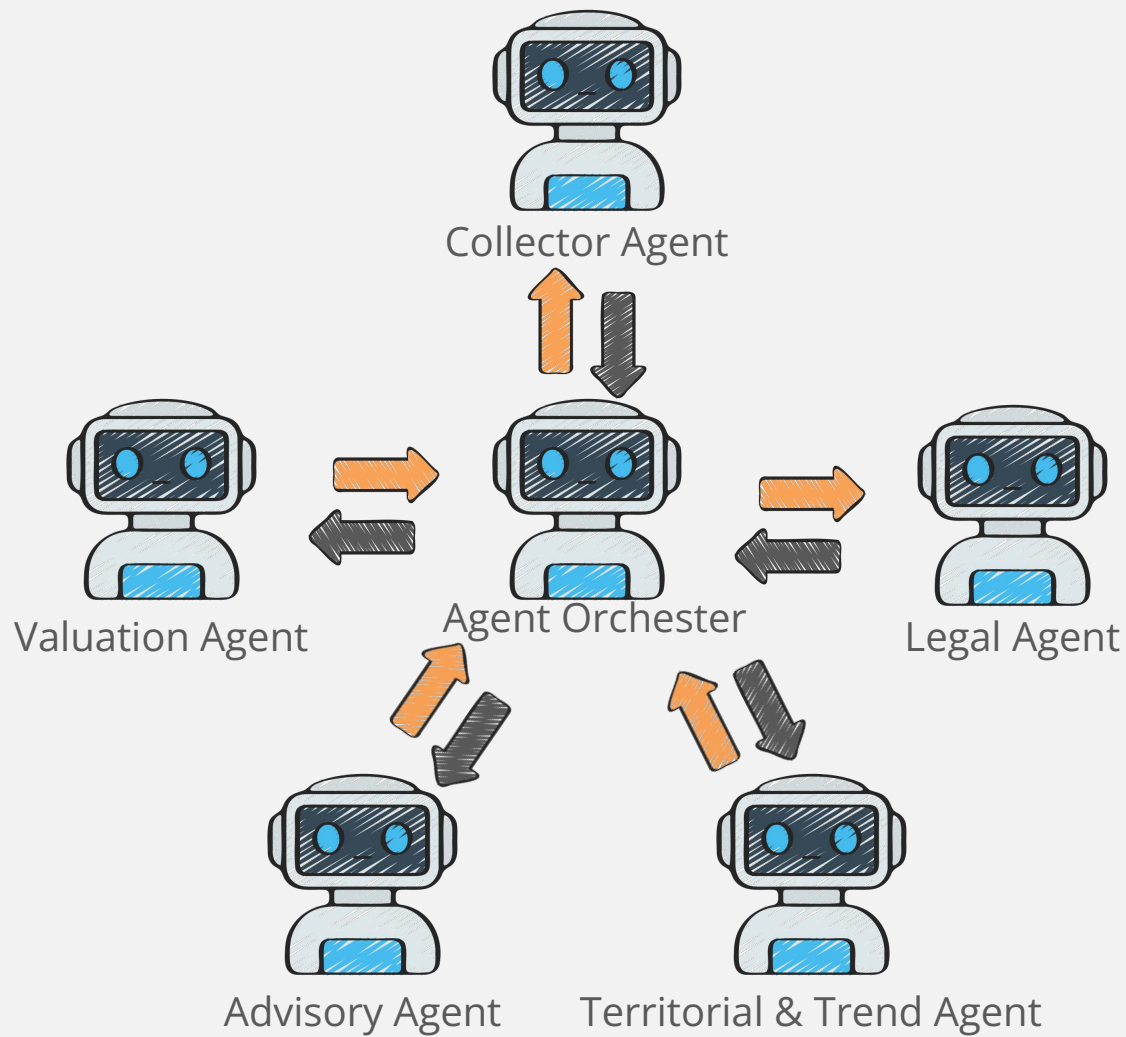
Advanced Data Collection
& Enrichment



Explainable &
Responsible AI

05. Business Objectives (BO) & Data Science Objectives (DSO)

Solution



BO (Business Objective)	DSO (Data Science Objectives)
BO1 — Improve Market Reliability Increase transparency and quality of real-estate listings.	DSO1: Automate multi-source data ingestion. DSO2: Structure and clean listings using NLP/LLMs. DSO3: Track, version, and orchestrate data pipelines.
BO2 — Understand Territorial Dynamics Identify spatio-temporal trends and emerging areas.	DSO1: Model market activity with time-series methods. DSO2: Perform spatial aggregation and mapping of listings. DSO3: Detect emerging zones and trigger alerts.
BO3 — Estimate Prices with Confidence Deliver accurate and explainable property valuations.	DSO1: Predict prices using supervised learning models. DSO2: Integrate temporal and territorial drivers. DSO3: Explain predictions using XAI techniques. DSO4: Detect under- and over-valued properties. DSO5: Expose valuations and explanations through APIs and visual analytics.
BO4 — Support Investment Decisions Optimize choices through personalized analytics.	DSO1: Build user profiles from preferences and behavior. DSO2: Simulate profitability and “what-if” scenarios. DSO3: Recommend properties and strategies via scoring models.
BO5 — Secure Legal Compliance Reduce regulatory risk and automate conformity checks.	DSO1: Model legal rules in a regulatory knowledge graph. DSO2: Continuously update regulatory data. DSO3: Compute an automated legal-risk score. DSO4: Detect non-compliance and issue alerts. DSO5: Orchestrate specialized agents for continuous monitoring.
BO6 — Make the Platform Operational Enable access through AI assistants and automated reporting.	DSO1: Allow interaction via a conversational assistant. DSO2: Automatically generate reports and summaries. DSO3: Ensure explainability and traceability of outputs.

Improve Market Reliability

BO



Increase transparency
and quality of real-estate
listings.



DSO



Automate
multi-source data
ingestion.



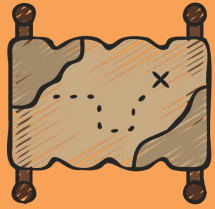
Structure and clean
listings using
NLP/LLMs.



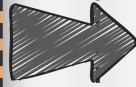
Track, version, and
orchestrate data
pipelines.

Understand Territorial Dynamics

BO



Identify spatio-temporal trends and emerging areas



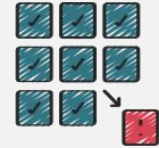
DSO



Model market activity with time-series methods



Perform spatial aggregation and mapping of listings



Detect emerging zones and trigger alerts

Estimate Prices with Confidence

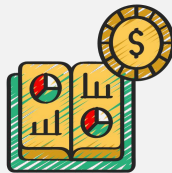
BO



Deliver accurate and explainable property valuations



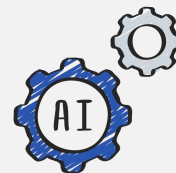
DSO



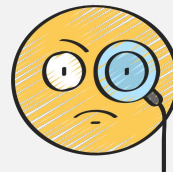
Predict prices using supervised learning models



Integrate temporal and territorial drivers



Explain predictions using XAI techniques



Detect under- and over-valued properties

Support Investment Decisions

BO



Optimize choices through
personalized analytics



DSO



Build user profiles
from preferences and
behavior



Simulate profitability
and "what-if"
scenarios.



Recommend
properties and
strategies using
scoring models

Secure Legal Compliance

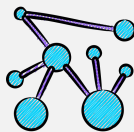
BO



Reduce regulatory risk
and automate conformity
checks.



DSO



Model legal rules in a
regulatory knowledge
graph



Continuously update
regulatory data.



Compute an
automated legal-risk
score.



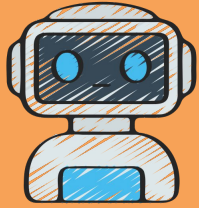
Detect
non-compliance and
issue alerts.



Orchestrate specialized
agents for continuous
monitoring.

Make the Platform Operational

BO



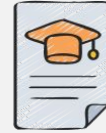
Enable access through AI assistants and automated reporting



DSO



Allow interaction via a conversational assistant



Automatically generate reports and summaries



Ensure explainability and traceability of outputs

06. Key Performance Indicators (KPIs)

Data Collection & Data Quality

Technical KPIs



Source coverage rate
(%)



Data collection
success rate

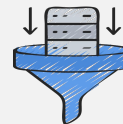


Automatically
structured listings
rate (%)

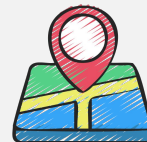
Business KPIs



Reliable listings rate
after quality control
(%)



Duplicate reduction
across platforms (%)



Geographic coverage
of the Tunisian
market (%)

Territorial & Temporal Market Dynamics

Technical KPIs



Correctly identified
emerging zones rate



Territorial trend
detection accuracy



Detection delay of
significant territorial
changes

Business KPIs



Number of emerging
zones identified and
monitored



Usage rate of
territorial maps



Adoption of territorial
insights by users

Price Estimation Models

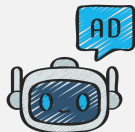
Technical KPIs



Average price
estimation error (MAE
/ RMSE)



Temporal stability of
model performance



Quality of model
explanations (XAI)

Business KPIs



Average gap between
asking price and
estimated price



User adoption of the
price estimation tool



Perceived trust in price
estimations

Legal Risk & Compliance

Technical KPIs



Regulatory risk score
accuracy



False positive / false
negative rate of legal
alerts



Compliance
evaluation response
time

Business KPIs



Reduction of risky
transactions



Early detection of
non-compliant listings



Perceived trust in
legal compliance

Decision Support & Recommendations

Technical KPIs



Recommendation
engine performance



Investment return
simulation accuracy



Relevant opportunity
alert rate

Business KPIs



Average investment
return improvement



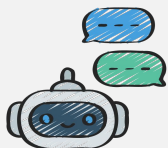
Decision time
reduction



Adoption rate of
recommendations

Intelligent Assistant

Technical KPIs



Relevant response
rate



Error / hallucination
rate



Assistant availability
(% uptime)

Business KPIs



Assistant adoption
rate



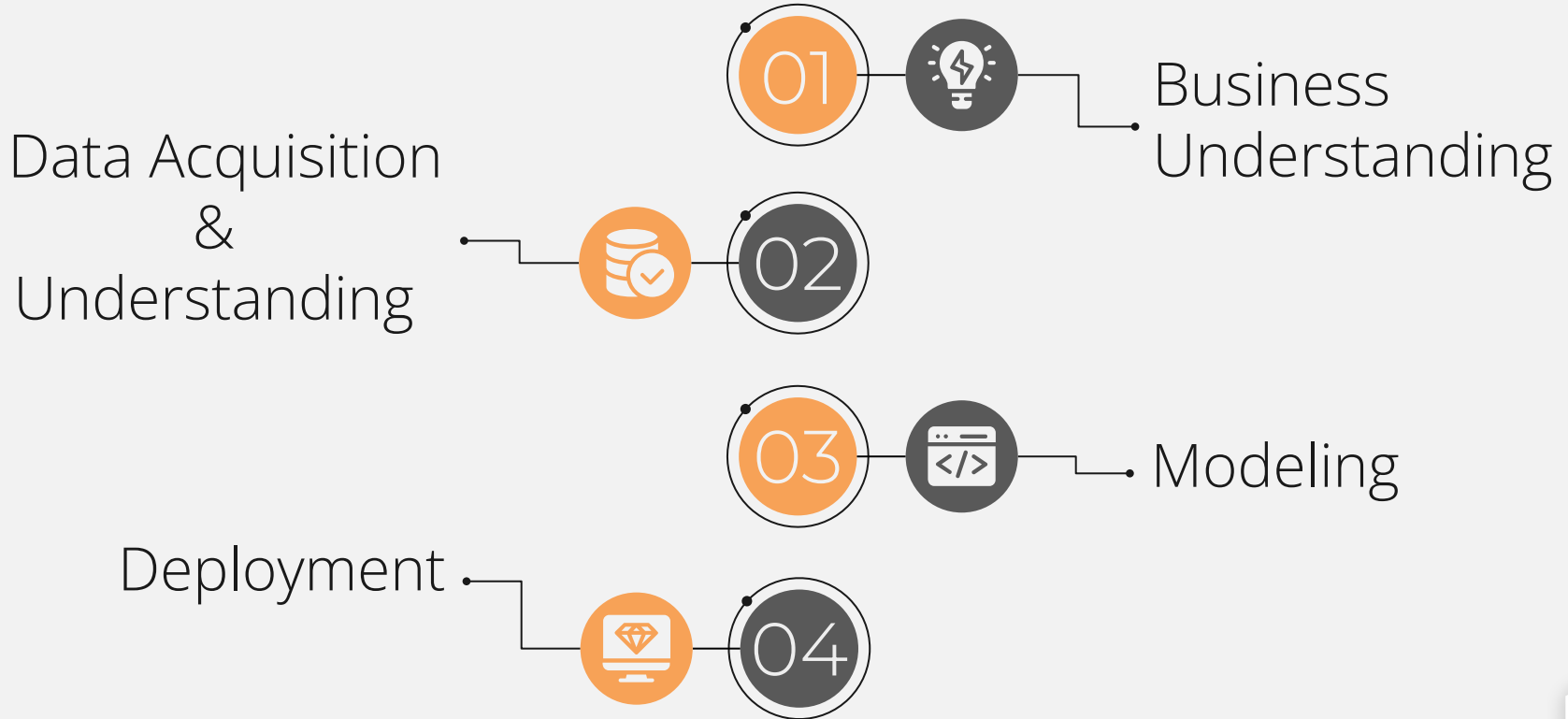
Manual analysis time
reduction



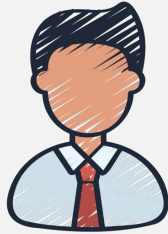
User satisfaction with
answer clarity

07. TDSP Methodology & Team Roles

TDSP Phases

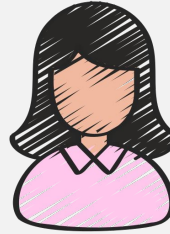


TDSP Roles



Project Manager

Bahar Wissem



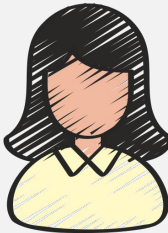
Project Lead

Bahar Wissal



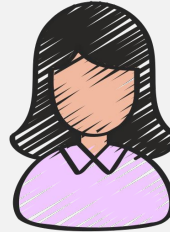
Data Scientist

Alaya Salma



Data Scientist

Mraieh Nourhen



Solution Architect

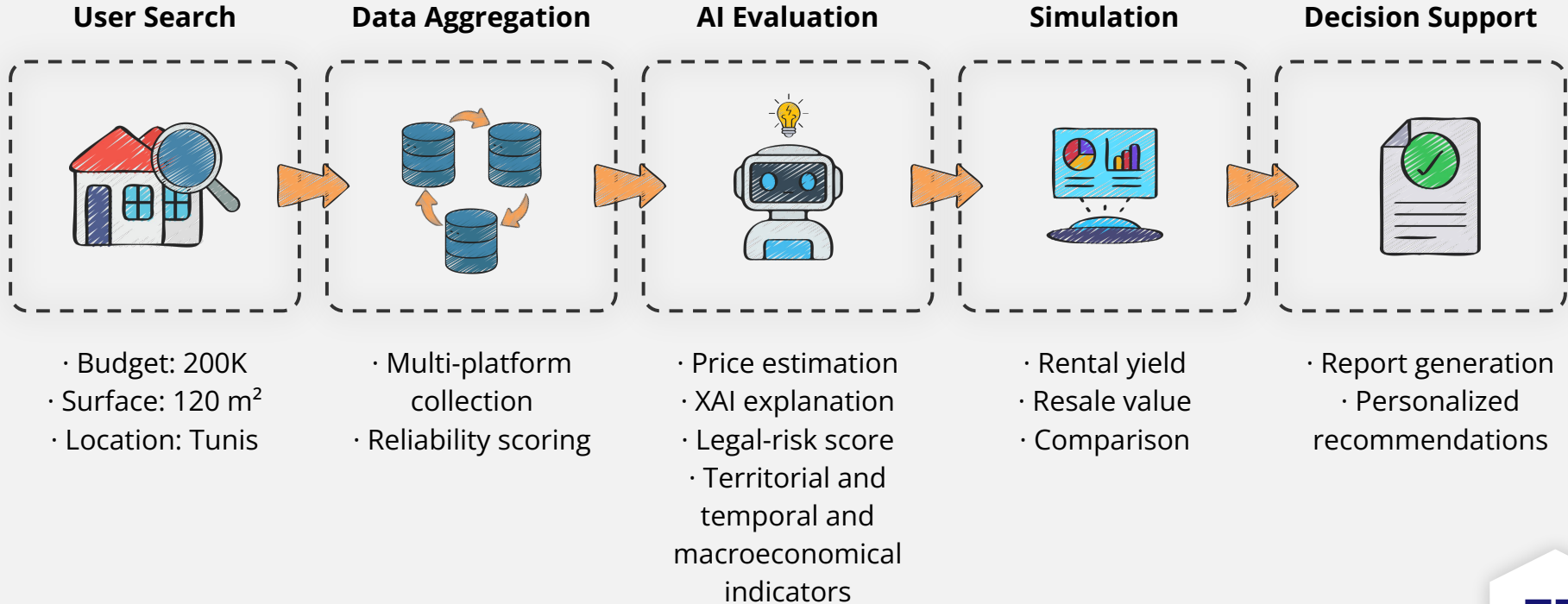
Ben Mahmoud Yosser



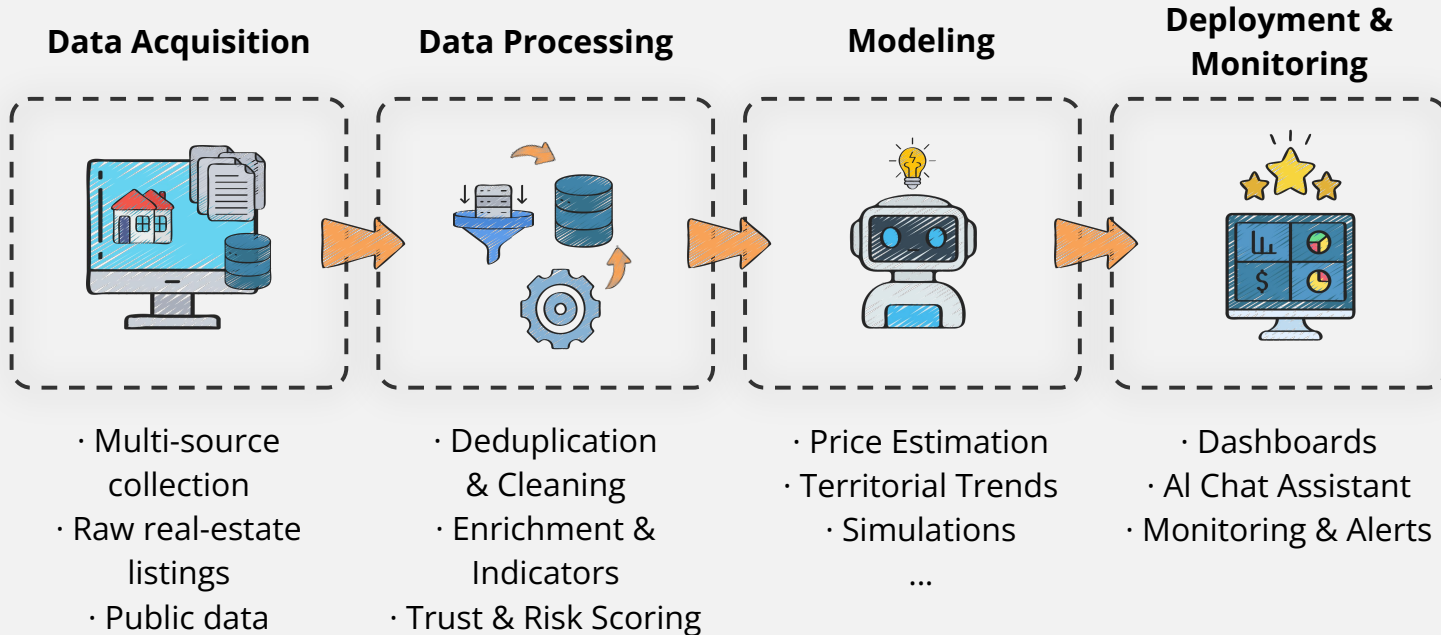
Business Analyst

Zaghouani Manar

Data Scenario – From Search to Decision



Project Workflow (TDSP Methodology)



Project Pipeline

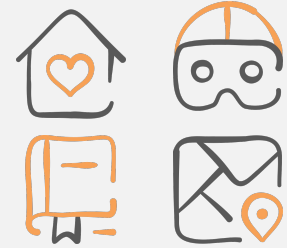


Input

Raw Data



End-to-End



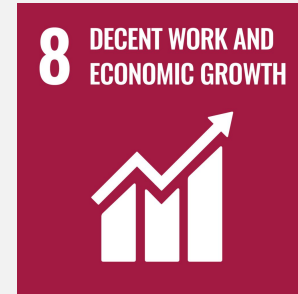
Output

- Interactive maps
- Explainable valuations
- Legal compliance
- Simulations
- Personalized recommendations
- Decision reports

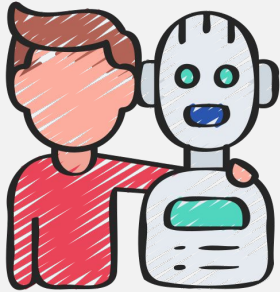


08. Ethics, Sustainability & Responsible AI

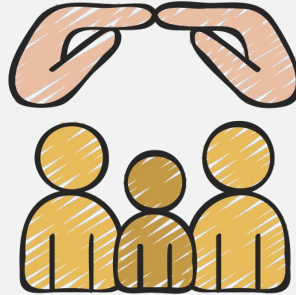
SDG



Ethical Principles



**Transparency &
Explainability**



Safety & Robustness



**Responsibility & Human
Oversight**



09. Next Steps: Data Understanding & Acquisition

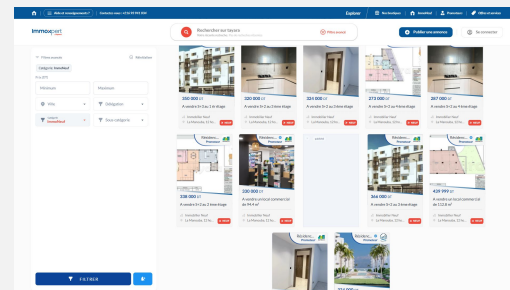
Data sources

- tayara
- mubawab
- Tunisie Anonce
- century21
- orpi
- Tecnocasa
- TPS Immobiliere
- Remax
- darkom
- verdar
- home in Tunisia

Data Understanding & Acquisition

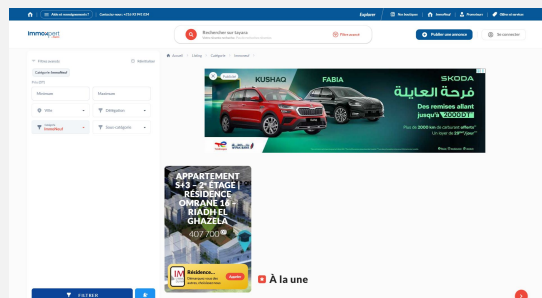
annombres_mubawab_clean.csv - Excel

ID	Titre	Type_Bien	Prix_TND	Ville	Quartier	Surface_m2	Pieces	Lien
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2	Bel appartem Bel			Sahloul	Sahloul			https://www.mubawab.tn/fr/pa/8284565/bel-appartement-s2-sahloul-4
3	APT Centre Vi APT			Tunis	Montfleury			https://www.mubawab.tn/fr/pa/8293596/apt-centre-ville-tunis
4	Appartement Appartement			Tunis	Montfleury			https://www.mubawab.tn/fr/pa/8297307/appartement-s2-haut-standing-%C3%A0-vendre-%E2%80%93-%C3%A0-buvidence-jeinyet-montfleury
5	S2 de 124,4m2			Ariana Ville	Les Jardins d'El Menzah 2			https://www.mubawab.tn/fr/pa/8248450/s2-de-124-4m%C2%82-avec-une-terrasse-de-43-70m%C2%82-%C3%A0-jardins-d%E2%80%99ne
6	Un prestigieux Un			La Marsa	La Perle du Lac 1			https://www.mubawab.tn/fr/pa/7961881/un-prstigieux-s3-%C3%A0-la-perle-du-lac-houses
7	54 172m² au 54			Le Kram	Les Jardins de Carthage			https://www.mubawab.tn/fr/pa/7928584/54-172m%C2%82-avec-une-terrasse-de-96m%C2%82-%C3%A0-jardin-de-carthage
8	10A Cidence / 10A Cidence			La Marsa	La Marsa			https://www.mubawab.tn/fr/pa/7914176/10a-cidence-%C3%A0-buvidence-%C3%A0-la-marsa
9	Vente Local c/Vente			Ezzahra	Ezzahra			https://www.mubawab.tn/fr/pa/7928845/vente-local-commercial-de-159-m%C2%82-%C3%A0-zahrat-el-medina-i-ezzahra
10	Vend appartei Vend			Hammam Sou	El Kantouli			https://www.mubawab.tn/fr/pa/8243281/vend-appartement-%C3%A0-el-kantouli-3-belles-chambres
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12	Local Comme Local			Chotrana 3	Chotrana 3			https://www.mubawab.tn/fr/pa/7941242/local-commercial-neuf-de-287-m%C2%82-%C3%A0-chotrana-3
13	Local Comme Local			Chotrana 3	Chotrana 3			https://www.mubawab.tn/fr/pa/7938052/local-commercial-neuf-de-320-m%C2%82-%C3%A0-chotrana-3
14	Vente Locaux Vente			La Soukra	Chotrana 3			https://www.mubawab.tn/fr/pa/7941270/vente-locaux-commerciaux-premium-%C3%A0-jjene-soukra-1
15	S3 Neuf de 17 S3			La Soukra	Chotrana 3			https://www.mubawab.tn/fr/pa/7938025/s3-neuf-de-177m%C2%82-%C3%A0-jjene-soukra-1-chotrana-3
16	A vendre S2 d A			La Soukra	Chotrana 1			https://www.mubawab.tn/fr/pa/8257532/a-vendre-s2-de-105-45m%C2%82-%C3%A0
17	Ville Jombok Villa			Hammamet	Hammamet Sud			https://www.mubawab.tn/fr/pa/8140132/villa-jombok
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19	Villa bulgari Villa			Hammamet	Hammamet			https://www.mubawab.tn/fr/pa/7947672/villa-bulgari
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24	Bel appartem Bel			La Marsa	Aouina			https://www.mubawab.tn/fr/pa/8185298/bel-appartement-%C3%A0-vendre-%C3%A0
25	RDC indEpe RDC			Monastir Ville	Skanes			https://www.mubawab.tn/fr/pa/8290994/rdc-ind%C3%A0pendant-et-standing
26	Appartement Appartement			La Soukra	La Soukra			https://www.mubawab.tn/fr/pa/8291252/appartement-s3
27	S2 vente A Ai S2			La Marsa	Ain Zaghouan			https://www.mubawab.tn/fr/pa/8293922/s2-vente-%C3%A0-ain-zaghouan
28	Villa de luxe Villa			Sfax Ouest	El Habib			https://www.mubawab.tn/fr/pa/8295702/villa-de-luxe-%C3%A0-vendre-dans-le-quart
29	Fonds de com Fonds			El Menzah	Mutuelle Ville			https://www.mubawab.tn/fr/pa/8295702/fonds-de-commerce-%C3%A2tisserie-%C3%A0



tayara_immoneuf - Excel

page,titre,prix,localisation,lien,image
1.À VENDRE – APPARTEMENT S+2 AU RDC RÉSIDENCE OMRANE 16 – RIADH EL GHAZELA,Promoteur,DT,https://www.tayara.tn/item/immobilier-neuf/ariana/ghazela/-vendre-appartement
1,Appartement S+3 – 4* étage Résidence Omrane 16 – Riadh El Ghazela,Promoteur,"Résidence Omrane 16
Promoteur",https://www.tayara.tn/item/immobilier-neuf/ariana/ghazela/appartement-s3-4-tage-rsidence-omrane-16-riadh-el-ghazela/69752728d921b01419d27d2c/https://www.tayara.
1,"Appartement S+3 RDC -Ref (F0-2B) – Résidence Omrane 16, Riadh El Ghazela",Promoteur,"Résidence Omrane 16
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1,À vendre – Appartement S+2 au RDC Résidence Omrane 16 – Riadh El Ghazela,Promoteur,"Résidence Omrane 16
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1,À vendre – Appartement S+2 au RDC Résidence Omrane 16 – Riadh El Ghazela,Promoteur,"Résidence Omrane 16
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1,À vendre – Appartement S+3 au 1er étage Résidence Omrane 16 – Riadh El Ghazela,Promoteur,"Résidence Omrane 16
Promoteur",https://www.tayara.tn/item/immobilier-neuf/ariana/ghazela/-vendre-appartement-s3-au-1er-tage-rsidence-omrane-16-riadh-el-ghazela/697545d0fea36ed612c8b3
1,Appartement S+3 – 6* étage Résidence Omrane 16 – Riadh El Ghazela,Promoteur,"Résidence Omrane 16
Promoteur",https://www.tayara.tn/item/immobilier-neuf/ariana/ghazela/appartement-s3-6-tage-rsidence-omrane-16-riadh-el-ghazela/697602c3fea36ed612c8ede5/https://w
1,Appartement S+3 – 3* étage Résidence Omrane 16 – Riadh El Ghazela,Promoteur,"Résidence Omrane 16



10. Conclusion



TuniEstate : See clearly...
before you invest.

Thanks for your
attention !



Etude de l'existant

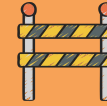


Avantages



- Rapports evolution des prix et les rendements par zone

Limites



- Approche descriptive
- Peu de personnalisation fine
- Pas de moteur conformité juridique

Inconvénients



- N'accompagnent pas l'utilisateur dans une décision précise sur un bien particulier.